

Data-Driven Neuroimaging Analysis

Nathaniel E. Helwig

Assistant Professor of Psychology and Statistics
University of Minnesota (Twin Cities)



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Outline of Talk

1) Component Analysis Methods:

- Bilinear Models
- Multiway Models
- Some analyses and results

2) Functional Modeling:

- Parametric Regression
- Nonparametric Regression
- Some analyses and results

Bilinear Model Form

Let $\mathbf{X} = \{x_{ij}\}_{I \times J}$ denote a data matrix.

- I voxels by J time points

A **bilinear model** with $R \leq \min(I, J)$ components has the form

$$\mathbf{X} = \mathbf{AB}' + \mathbf{E} \quad \Longleftrightarrow \quad x_{ij} = \sum_{r=1}^R a_{ir}b_{jr} + e_{ij}$$

where

- $\mathbf{A} = \{a_{ir}\}_{I \times R}$ are the voxel weights (spatial maps)
- $\mathbf{B} = \{b_{jr}\}_{J \times R}$ are the time weights (time maps)
- $\mathbf{E} = \{e_{ij}\}_{I \times J}$ are the error terms

Rotational Indeterminacy Problem

Suppose that $\mathbf{R}$ is an $R \times R$ orthogonal rotation matrix.

- $\mathbf{R}'\mathbf{R} = \mathbf{R}\mathbf{R}' = \mathbf{I}$ (identity matrix)

Rotational indeterminacy problem of the bilinear model:

$$\mathbf{AB}' = \tilde{\mathbf{A}}\tilde{\mathbf{B}}'$$

where $\tilde{\mathbf{A}} = \mathbf{AR}$ and $\tilde{\mathbf{B}} = \mathbf{BR}$.

Need to make some (non-data-driven) assumptions to solve the rotational indeterminacy.

Principal Components and Independent Components

Principal Component Analysis Assumptions:

- Columns of **A** are orthogonal; $d_1 \geq \dots \geq d_R$ where $\mathbf{D} = \text{diag}(\mathbf{A}'\mathbf{A})$
- Columns of **B** are orthonormal
- See Pearson (1901) and Hotelling (1933)

Independent Component Analysis Assumptions:

- Columns of **A** are orthogonal and statistically independent
- Columns of **B** are orthonormal
- See Comon (1994) and De Lathauwer et al. (2000)

Historical Development of Multiway Models

Cattell's (1944) **Principle of Parallel Proportional Profiles:**

- Attempted to resolve rotational indeterminacy in bilinear model
- “Real” components should have an invariant structure across different conditions, and should vary proportionally depending on the relative importance of each component in each condition

Some popular multiway models:

- Tucker's (1966) 3-way factor model (Tucker3 Model)
- Harshman's (1970) Parallel Factors (PARAFAC) Model
- Carroll and Chang's (1970) Canonical Decomposition Approach

PARAFAC/CANDECOMP Model Form

Let $\mathbf{X} = \{x_{ijk}\}_{I \times J \times K}$ denote a three-way data array.

- x_{ijk} is signal from i -th voxel at j -th time point of k -th subject

A **trilinear model** has the form

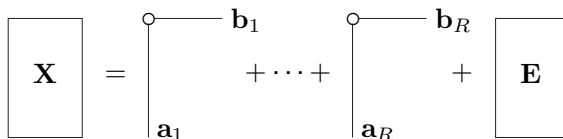
$$x_{ijk} = \sum_{r=1}^R a_{ir} b_{jr} c_{kr} + e_{ijk}$$

where

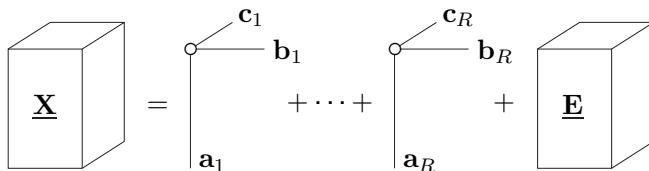
- $\mathbf{A} = \{a_{ir}\}_{I \times R}$ are the Mode A (voxel) weights
- $\mathbf{B} = \{b_{jr}\}_{J \times R}$ are the Mode B (time) weights
- $\mathbf{C} = \{c_{kr}\}_{K \times R}$ are the Mode C (subject) weights
- $\mathbf{E} = \{e_{ijk}\}_{I \times J \times K}$ is the 3-way error array

Visualization of PCA and Parafac (Helwig et al., 2012)

a) PCA: $\mathbf{X} = \sum_{r=1}^R \mathbf{a}_r \circ \mathbf{b}_r + \mathbf{E}$



b) Parafac: $\underline{\mathbf{X}} = \sum_{r=1}^R \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r + \underline{\mathbf{E}}$



Parafac Uniqueness (Intrinsic Axis Property)

Parafac is popular due to its **essential uniqueness** property.

- First suggested by Harshman (1970, 1972)
- Sufficient condition provided by Kruskal (1977)

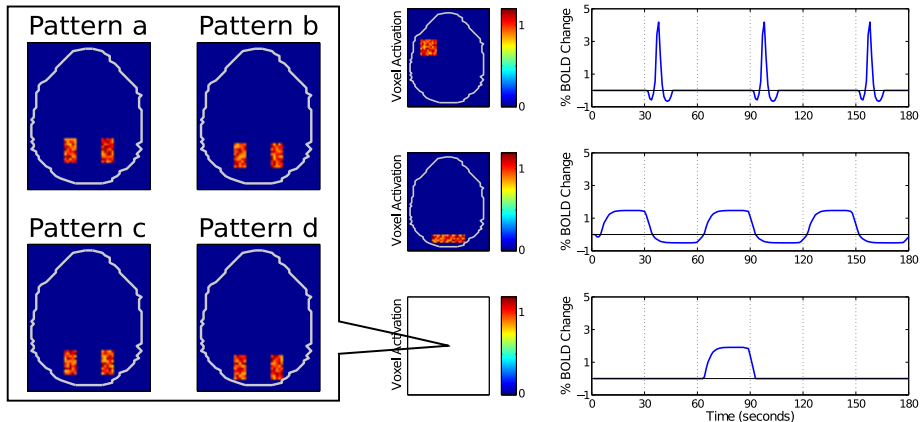
Kruskal's sufficient condition for Parafac essential uniqueness:

- Define the **k -rank (Kruskal-rank)** of $\mathbf{Y}$ to be the largest integer k such that every subset of k columns of $\mathbf{Y}$ is linearly independent
- Suppose $(\mathbf{A}, \mathbf{B}, \mathbf{C})$ is a rank- R Parafac solution, and suppose that $k_{\mathbf{A}} + k_{\mathbf{B}} + k_{\mathbf{C}} \geq 2R + 2$, where $k_{(\cdot)}$ is k -rank of denoted matrix
- If $(\tilde{\mathbf{A}}, \tilde{\mathbf{B}}, \tilde{\mathbf{C}})$ is rank- R solution with the same error array $\mathbf{E}$, then

$$\tilde{\mathbf{A}} = \mathbf{A}\mathbf{P}\mathbf{D}_{\mathbf{A}}, \quad \tilde{\mathbf{B}} = \mathbf{B}\mathbf{P}\mathbf{D}_{\mathbf{B}}, \quad \tilde{\mathbf{C}} = \mathbf{C}\mathbf{P}\mathbf{D}_{\mathbf{C}}$$

where $\mathbf{P}$ is a permutation matrix and $(\mathbf{D}_{\mathbf{A}}, \mathbf{D}_{\mathbf{B}}, \mathbf{D}_{\mathbf{C}})$ are diagonal scaling matrices such that $\mathbf{D}_{\mathbf{A}}\mathbf{D}_{\mathbf{B}}\mathbf{D}_{\mathbf{C}} = \mathbf{I}_R$ (identity matrix)

Some Simulated fMRI Data (Helwig & Hong, 2013)



Spatial pattern of activity differs across four patterns:

- (a) No overlapping activity
- (b) 10% overlapping activity
- (c) 30% overlapping activity
- (d) 50% overlapping activity

Parafac versus TPICA for fMRI Data Analysis

TPICA: Tensor Probabilistic Independent Component Analysis

- Combines ideas of ICA and Parafac
- Assumes spatial maps are independent

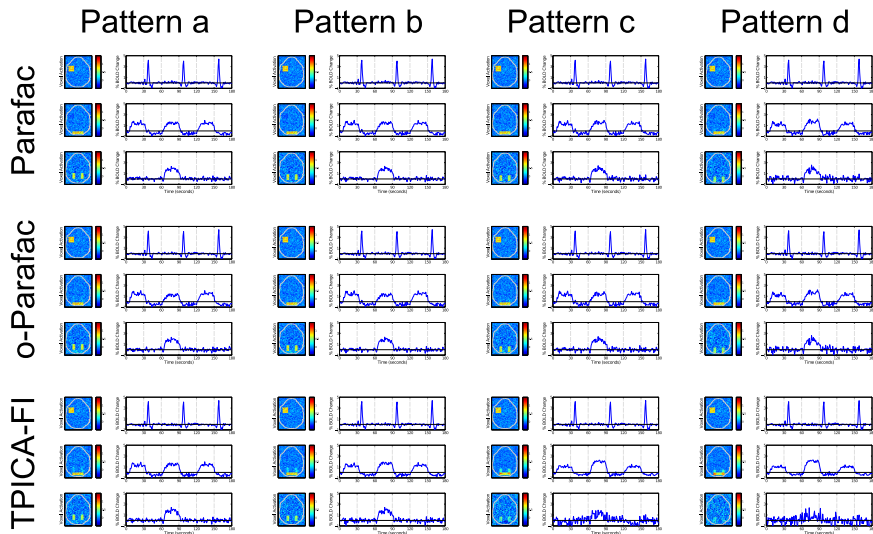
Comparing TPICA and Parafac for fMRI:

- If independence assumption is correct. . .
 - TPICA outperforms Parafac at small SNR
 - Orthogonal-Parafac performs as well as TPICA at small SNR
- If independence assumption is incorrect. . .
 - TPICA produces noticeable bias for all SNR
 - Orthogonal-Parafac produces smaller bias than TPICA

See Helwig & Hong (2013) for thorough critique of TPICA

Parafac versus TPICA (SNR=0.10)

Figure 4 from Helwig and Hong (2013):



Multiple Linear Regression Model

The **multiple linear regression** model has the form

$$y_i = b_0 + \sum_{j=1}^p b_j x_{ij} + e_i \quad \Longleftrightarrow \quad \mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{e}$$

for $i \in \{1, \dots, n\}$ where

- $y_i \in \mathbb{R}$ is the real-valued **response** for the i -th observation
- $b_0 \in \mathbb{R}$ is the regression **intercept**
- $b_j \in \mathbb{R}$ is the j -th predictor's regression **slope**
- $x_{ij} \in \mathbb{R}$ is the j -th **predictor** for the i -th observation
- $e_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ is Gaussian **measurement error**

Ordinary Least Squares and Inference

Typically minimize the ordinary least squares (OLS) problem

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where $\hat{y}_i = \hat{b}_0 + \sum_{j=1}^p \hat{b}_j x_{ij}$ is the fitted value.

$\hat{\mathbf{b}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ and from assumptions $\hat{\mathbf{b}} \sim N(\mathbf{b}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$.

- Inference follows from the normality assumption

Limitations of Parametric Regression

MLR parameters have conditional interpretation.

- b_j gives effect of X_j holding other X 's constant
- Hard to interpret if X_j variables are correlated

MLR assumes effects have known parametric (e.g., linear) relationship.

- With real data, one rarely knows the parametric form
- Typically we collect data because we want to discover the form of the relationships between variables

Nonparametric Regression Model

Assume observed data follow a **nonparametric regression** model

$$y_i = \eta(\mathbf{x}_i) + e_i$$

where

- $y_i \in \mathbb{R}$ is observed response for the i -th observation
- $\mathbf{x}_i \equiv (x_{i1}, \dots, x_{ip})'$ is the i -th observation's known predictor vector
- η is an unknown smooth function
- $e_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$ is unobserved measurement error

Goal is to estimate the unknown function η from the noisy data.

Model Building and Penalized Least Squares

You can include any combination of effects in the model. For example, with two predictors $\mathbf{x} = (x_1, x_2)$ we could fit two possible models:

- Additive: $\eta(\mathbf{x}) = \eta_0 + \eta_1(x_1) + \eta_2(x_2)$
- Interaction: $\eta(\mathbf{x}) = \eta_0 + \eta_1(x_1) + \eta_2(x_2) + \eta_{12}(x_1, x_2)$

Once η is specified, minimize penalized least squares functional

$$\frac{1}{n} \sum_{i=1}^n (y_i - \eta(\mathbf{x}_i))^2 + \lambda J(\eta)$$

where

- $J(\eta) > 0$ is a **penalty functional** quantifying roughness of η
- $\lambda > 0$ is a **smoothing parameter** controlling influence of J

Smoothing Spline Analysis of Variance

Estimates η in tensor product reproducing kernel Hilbert space $\mathcal{H}$.

Two-way SSANOVA useful for neuroimaging data analysis:

$$\eta(\mathbf{x}) = \eta_0 + \eta_1(\text{space}) + \eta_2(\text{time}) + \eta_{12}(\text{space, time})$$

Three-way SSANOVA useful for neuroimaging data analysis:

$$\begin{aligned}\eta(\mathbf{x}) = & \eta_0 + \eta_1(\text{space}) + \eta_2(\text{time}) + \eta_3(\text{group}) \\ & + \eta_{12}(\text{space, time}) + \eta_{13}(\text{space, group}) \\ & + \eta_{23}(\text{time, group}) + \eta_{123}(\text{space, time, group})\end{aligned}$$

Visual Stimulus ERP Study

Data from UCI Machine Learning Repository; originally collected by **Henri Begleiter** at the Neurodynamics Laboratory at SUNY Brooklyn.

Experiment Details:

- Data from 20 subjects
 - 10 alcoholics
 - 10 controls
- 64-channel EEG cap
 - Ground: Nz
 - Reference: Cz
- Snodgrass & Vanderwart
 - Set of 260 pictures
- 10 trials sampled at 256 Hz for one second after stimulus

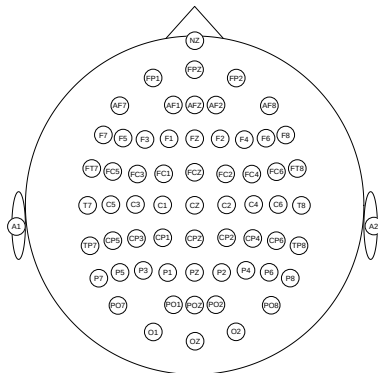


Figure: The 64-channel EEG cap.

Data

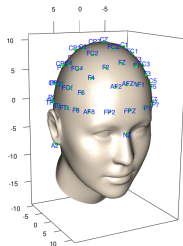
For the multi-electrode (spatial) analysis, I focus on EEG activity at all 61 electrodes from controls and alcoholics

- Electrode: $x_{i1} \in \{1, 2, 3, 4, \dots, 61\} \rightarrow \mathbb{R}^3$
- Time: $x_{i2} \in \{0, 1, 2, 3, \dots, 255\} \rightarrow [0, 1]$
- Group: $x_{i3} \in \{\text{control}, \text{alcoholic}\} \rightarrow \{1, 2\}$

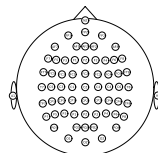
Map each electrode to a 3-dimensional coordinate in Euclidean space.



Registration



Projection



Analyses

Three-way SSANOVA on $\mathcal{X} \equiv \mathbb{R}^3 \times [0, 1] \times \{1, 2\}$

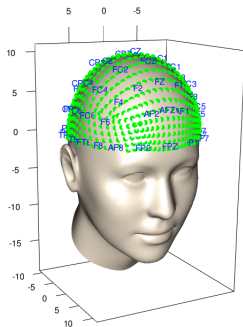
- Thin-plate spline for electrode effect
- Cubic spline for time effect
- Nominal spline for group effect

I analyzed data from all subjects and electrodes, so that $n = 3123200$ (20 subjects $\times$ 256 time points $\times$ 61 electrodes $\times$ 10 trials).

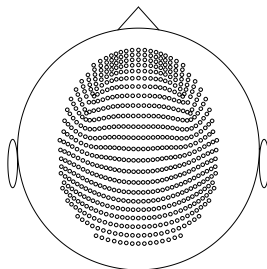
Examined the solution using $q = 1520$ bin-sampled knots.

Predicted ERPs

Can predict function in 3D space with dense cap (633 electrodes), and then project results to 2D space for visualization.



Projection



Useful for visualizing **predicted ERPs and ERP differences**.

Some R Packages

Some R packages for multiway modeling:

- `ThreeWay`: Three-way component analysis. (Ferraro et al., 2014)
- `multiway`: Component Models for Multiway Data. (Helwig, 2015b)

Some R packages for nonparametric regression:

- `gss`: General Smoothing Splines. (Gu, 2014)
- `mgcv`: Mixed GAM Computation Vehicle. (Wood, 2015)
- `bigsplines`: Smoothing Splines for Large Samples. (Helwig, 2015a)

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